

Practical 7

17th September 2026

Questions

1. Compute the amino acid composition of the following sequences. Provide the output as a table of amino acid percentage values for each sequence and comment on the results.

1. RATPTRWPVGCFNRPWTKWSYDEALDGIKAAGYAWTGLLTASKPSLHH
ATATPEYLAALKQKSRHAA
2. AAAVMMGLAAIGAAIGIGILGGKFLEGAARQPDLIPLLRTQFFIVMGLVD
AIPMIAVGLGLYVMFAVA
3. AADVSAAVGATGQSGMTYRLGLSWDWDKSWWQTSTGRLTGYWDAGY
TYWEGGDEGAGKHSLFAPVFVYEFAGDSIKPFIEAGIGVAAFSGTRVGD
QNLGSSLNFEDRIGAGLKFANGQSVGVRAIHYSNAGLKQPNDGIESYSLF
YKIPI

2. Assume the molecular weights of the 20 amino acid residues as given below. Compute the molecular weight of the three sequences given in question 1.

Ala: 85 Cys: 115 Asp: 130 Glu: 145 Phe: 160 Gly: 70 Trp: 200
His: 150 Ile: 125 Lys: 145 Leu: 125 Met: 143 Asn: 130 Tyr: 175
Pro: 110 Gln: 140 Arg: 170 Ser: 100 Thr: 115 Val: 110

3. The amino acid composition of a standard set of Group A (first value) and Group B (second value) proteins are given below. Identify whether the given sequences in Question 1 belong to Group A or Group B and write your answer.

Ala: 8.47, 8.95 Asp: 5.97, 5.91 Cys: 1.39, 0.47 Glu: 6.32, 4.78 Thr: 5.79, 6.54
Phe: 3.91, 3.68 Gly: 7.82, 8.54 His: 2.26, 1.25 Ile: 5.71, 4.77 Val: 7.02, 6.76
Lys: 5.76, 4.93 Leu: 8.48, 8.78 Met: 2.21, 1.56 Asn: 4.54, 5.74 Trp: 1.44, 1.24
Pro: 4.63, 3.74 Gln: 3.82, 4.75 Arg: 4.93, 5.24 Ser: 5.94, 8.05 Tyr: 3.58, 4.13

4. Compute the residue pair preference for the three sequences given in question 1. The required output is a 20x20 table showing the pair preferences (a) $[N_{ij} \cdot 100 / (N_i + N_j)]$, (b) $[N_{ij} \cdot 100 / (N - 1)]$ and (c) $[N_{ij} \cdot 100 / (N_i \cdot N_j)]$. List the top 10 preferred residue pairs from each of the three pair-preferences.
5. Compute average hydrophobicity (Hgm), Helical contact area (Ca) and Total non-bonded energy (Et) for the sequences in Q1 and comment on the results. (Refer https://www.iitm.ac.in/bioinfo/fold_rate/prop_orig.html for the properties).

Deadline: 23rd September 2026